

AI & ML TASK 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Project: California Housing Price Prediction

Overfitting Analysis & Cross-Validation

Overfitting Detection (Train vs Test RMSE)

- Decision Tree (untuned) showed the largest train-test gap → clear sign of overfitting on unrestricted depth
- Random Forest showed a much smaller gap → ensemble averaging naturally controls variance
- Linear & Ridge Regression showed no overfitting — instead they underfit, unable to capture non-linear patterns

Model	Train RMSE	Test RMSE	Gap
Decision Tree	Low	0.7021	Large
Random Forest	Low	0.5059	Small
Ridge / Linear	Moderate	0.7456	≈None (underfit)

Why Cross-Validation Was Used

- A single train-test split can give a misleading performance estimate
- 5-fold cross-validation was used to get a stable, reliable RMSE estimate for each model
- CV results guided GridSearchCV tuning instead of relying on one split

5-Fold CV RMSE (Mean)	
Model	Mean CV RMSE
Random Forest	≈ 0.50 (lowest, most stable)
Decision Tree	≈ 0.68 (higher variance)
Ridge / Linear	≈ 0.75 (consistent, limited)

Hyperparameter Tuning Approach

Method: GridSearchCV with 5-fold cross-validation, scored on negative RMSE

Ridge Regression

GRID SEARCHED

Grid: $\alpha \in [0.01, 0.1, 1.0, 10.0, 50.0, 100.0]$

BEST PARAMS FOUND

Best alpha selected via grid search — no meaningful change vs. default

Decision Tree

GRID SEARCHED

Grid: max_depth [3, 5, 7, 10]
min_samples_split [2, 5, 10]

BEST PARAMS FOUND

Best combination reduced depth → controlled overfitting

Random Forest

GRID SEARCHED

Grid: n_estimators [100, 200]
max_depth [None, 10, 20], min_samples_split [2, 5]

BEST PARAMS FOUND

Best combination slightly increased tree count / adjusted split threshold

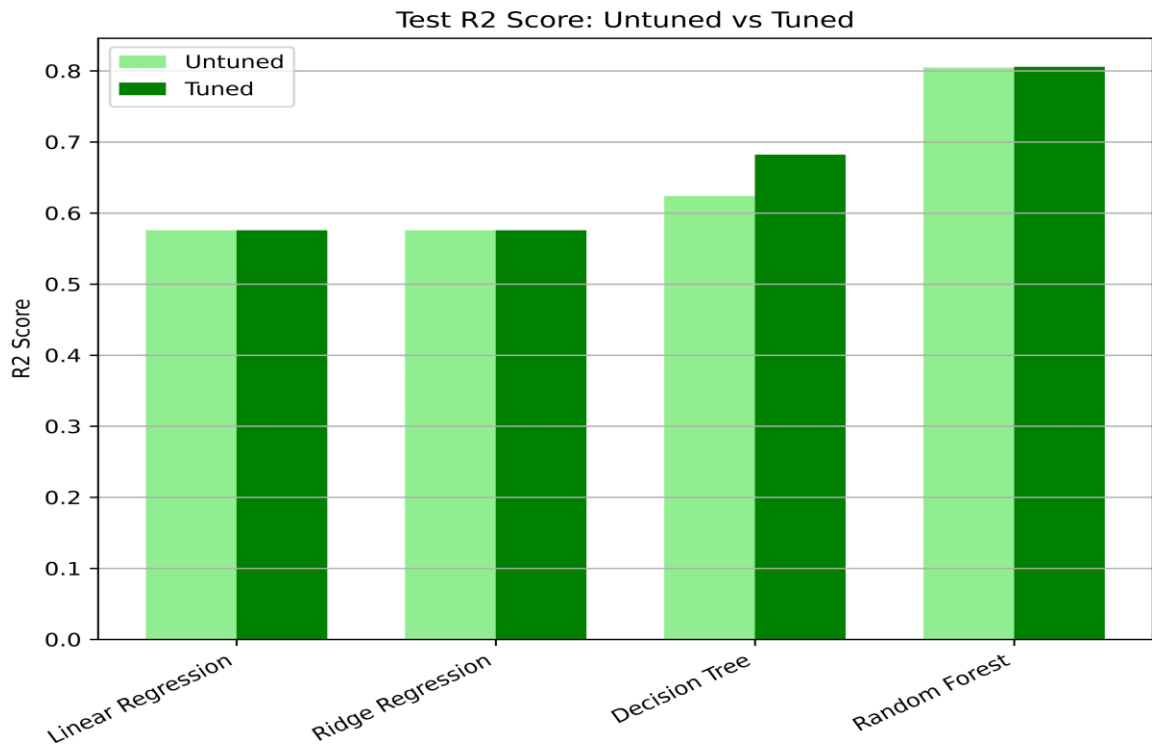
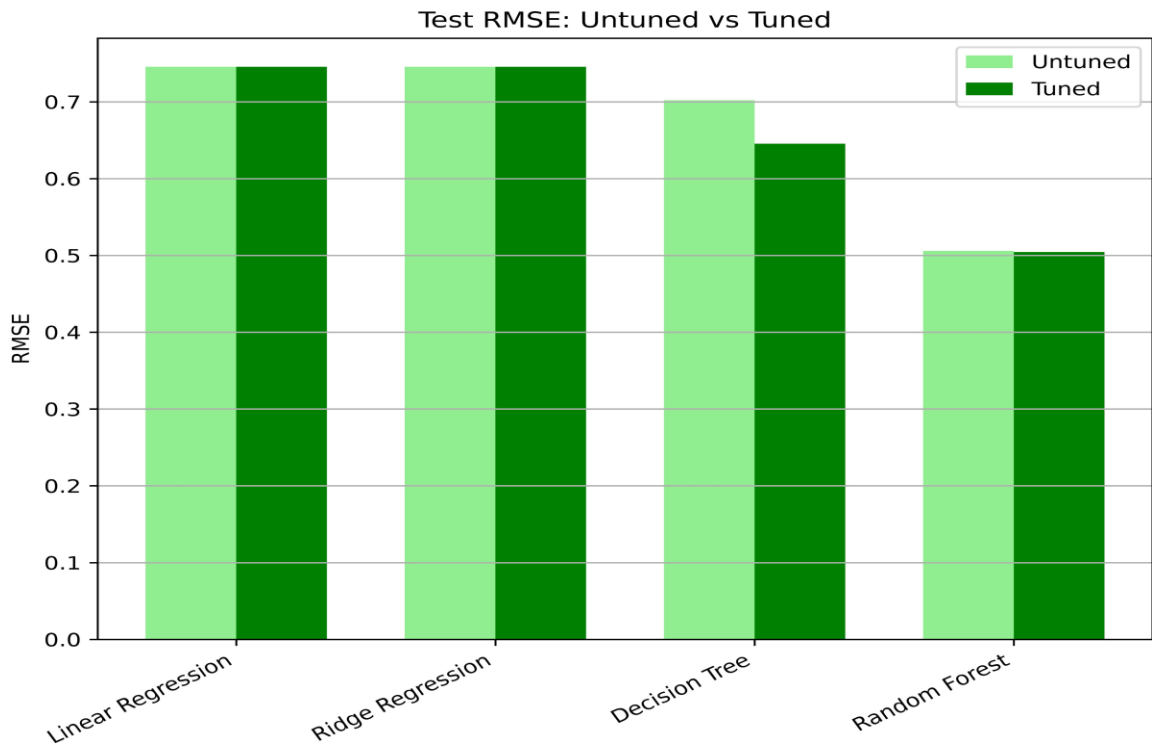
Note: Linear Regression has no hyperparameters and could not be tuned — included in comparisons as a fixed baseline.

Model Comparison: Untuned vs Tuned

	Model	Untuned RMSE	Tuned RMSE	Untuned R2	Tuned R2
3	Random Forest	0.5059	0.5045	0.8047	0.8058
2	Decision Tree	0.7021	0.6454	0.6238	0.6822
1	Ridge Regression	0.7456	0.7456	0.5758	0.5758
0	Linear Regression	0.7456	0.7456	0.5758	0.5758

Key Observation

Tuning helped Decision Tree most (R^2 0.6238 $\rightarrow$ 0.6822). Random Forest improved only marginally — it was already well-regularized through ensembling. Ridge showed no change, confirming its ceiling isn't tuning-related.



Final Model Selection & Key Learnings

FINAL MODEL SELECTED

Tuned Random Forest

0.5045

Test RMSE

0.8058

Test R²

WHY THIS MODEL

- Combines strong accuracy with stable generalization, validated via 5-fold CV
- Small train-test gap confirms it isn't overfitting despite its complexity
- Outperforms all baseline and tuned linear/tree alternatives

TRADE-OFFS

- More computationally expensive and less interpretable than Linear/Ridge Regression
- Tuning gain over untuned Random Forest was small — most strength already came from the ensembling structure itself

KEY TAKEAWAYS

- A single train-test split can hide overfitting — cross-validation gives a more trustworthy estimate
- Hyperparameter tuning helps most where a model is prone to overfitting (e.g., Decision Tree), and less where it's already well-regularized (Random Forest) or structurally limited (Ridge)
- Model selection should balance accuracy, generalization, and interpretability — not accuracy alone